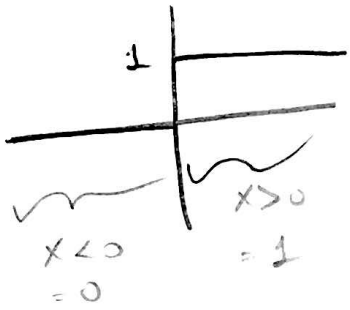
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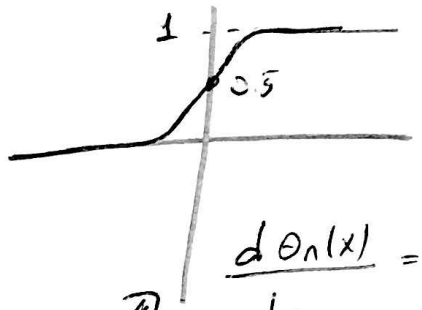
4) $\theta(x) = \int_{-\infty}^{x} \delta(x')\,dx'$



Heaviside "function"

Unit "function"

by integral theorem: $\frac{d\,\theta(x)}{dx} = \frac{dU(x)}{dx} = \delta(x)$



$\frac{d\,\theta_n(x)}{dx} = \delta_n(x)$

Hoca said "it's a good question."

Dirac Delta "Function" in 3D

$$\nabla\left(\frac{1}{r}\right) = \nabla\left(x^2+y^2+z^2\right)^{\frac{-1}{2}} = \frac{-\hat{r}}{r^2}, \qquad \int_V \nabla\cdot\left(\nabla\frac{1}{r}\right) d\tau$$

$$= -\int \nabla\cdot\left(\frac{\hat{r}}{r^2}\right) d\tau =$$

$$\nabla^2\frac{1}{r} = -4\pi\,\delta(\vec{r})$$

$$= \begin{cases} -4\pi, & V \text{ includes origin} \\ 0, & \text{otherwise.} \end{cases}$$

$$= -4\pi\,\delta(x)\,\delta(y)\,\delta(z)$$

looks like Dirac delta function.

$$\int_V \nabla^2\frac{1}{r}\,d\tau = -4\pi\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \delta(x)\,\delta(y)\,\delta(z)\,\underbrace{dx\,dy\,dz}_{d\tau}$$

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$$\int f(\vec{r}-\vec{r}_0) f(\vec{r}) d\tau = f(\vec{r}_0)$$

$$\nabla^2 \frac{1}{r} = -4\pi f(\vec{r})$$

$$\nabla \cdot \left(\frac{\hat{r}}{r^2}\right) = 4\pi f(\vec{r})$$

Complex Numbers — 2. Ders

$$az^2 + bz + c = 0$$

$$z = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$$

$$d^2 \equiv b^2 - 4ac$$

if $d^2 < 0$, "they" used to say this is not physical, and ignore the solutions. Some time z does not have to be physical.

$$i \equiv \sqrt{-1} \quad \text{and} \quad i^2 = -1$$

$$\sqrt{-2}\sqrt{-8} \stackrel{?}{=} \sqrt{(-2)\cdot(-8)}$$

$i\sqrt{2} \quad 2i\sqrt{2}$

$$4i^2 = \sqrt{16}$$

$$-4 \neq 4$$ *

$$z^2 - 2z + 2 = 0$$

$$z = 1 \pm i$$

$z = a + bi$ rectangular representation. both are real

$$Re\{z\} = a$$

$$Im\{z\} = b$$

$\mathbb{Z}$

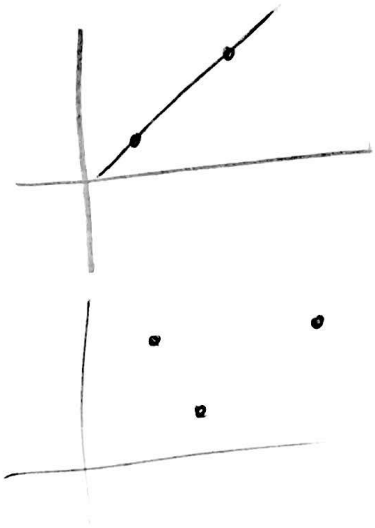
$$x + iy = r[\cos\theta + i\sin\theta]$$

x, y $\quad$ r, θ

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• Expanding Functions



can find a line equation pass through the given points

$$y = a_0 + a_1 x$$

2 nd degree unique polynomial pass through given 3 points

$$y = a_0 + a_1 x + a_2 x^2$$

In general we can find unique n^{th} degree polynomial that perfectly fits $n+1$ points

A generalized smooth (= undykied") function has ∞ points $\Rightarrow \infty$ th degree polynomials.

The question is: How we can find a_n $n=0 \dots \infty$ for such polynomial that corresponds to $f(x)$

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 - \dots$$

Find taylor exp. series for (i) $\sin x$ (ii) $\cos x$ iii) e^x

$$f(0) = a_0$$

$$f'(0) = a_1$$

$$f''(x) = 2a^2 + 2 \cdot 3\, a_3 x + \cdot 3 \cdot 4\, a_4 x^2 + \dots \qquad \frac{f''(0)}{2} = a_2$$

$$f^{(3)}(x) = 2 \cdot 3\, a_3 + 2 \cdot 3 \cdot 4\, a_4 x + \cdot 3 \cdot 4 \cdot 5\, a_5 x^2 = \frac{f^{(3)}(0)}{2 \cdot 3} = a_3$$

$$a_n = \frac{f^{(n)}(0)}{n!} \Rightarrow \boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0) \cdot x^n}{n!}}$$

Taylor's expansion

(Maclaurin expansion)